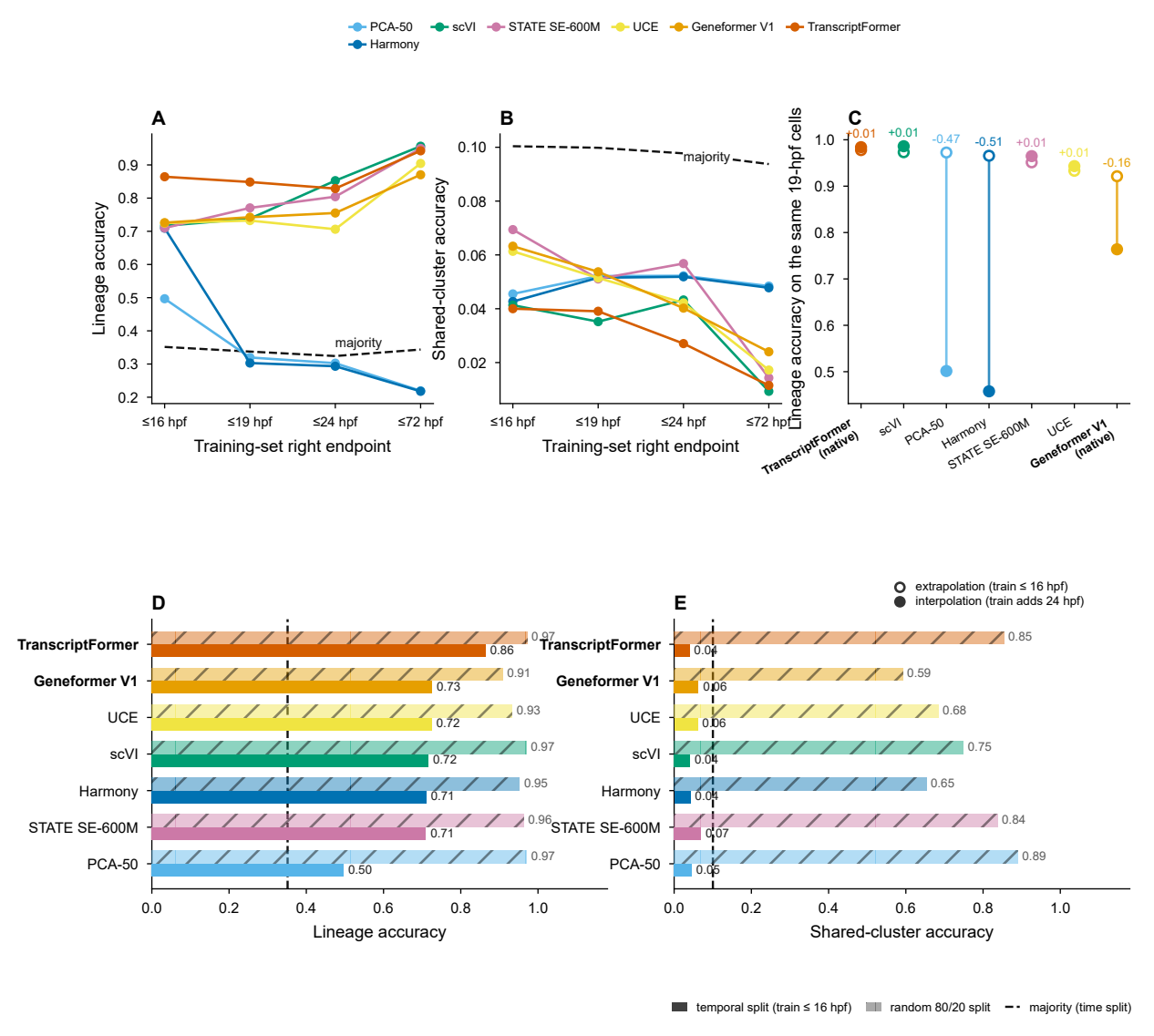

PCA-50
scVI
STATE SE-600M
UCE
Geneformer V1
TranscriptFormer
Harmony
A
Lineage accuracy
majority
≤16 hpf
≤19 hpf
≤24 hpf
≤72 hpf
Training-set right endpoint
B
Shared-cluster accuracy
majority
Training-set right endpoint
C
Lineage accuracy on the same 19-hpf cells
+0.01
+0.01
-0.47
-0.51
+0.01
+0.01
-0.16
TranscriptFormer (native)
scVI
PCA-50
Harmony
STATE SE-600M
UCE
Geneformer V1 (native)
D
TranscriptFormer
Geneformer V1
UCE
scVI
Harmony
STATE SE-600M
PCA-50
0.86
0.97
0.73
0.91
0.72
0.93
0.72
0.97
0.71
0.95
0.71
0.96
0.50
0.97
Lineage accuracy
E
extrapolation (train ≤ 16 hpf)
interpolation (train adds 24 hpf)
0.04
0.85
0.06
0.59
0.06
0.68
0.04
0.75
0.04
0.65
0.07
0.84
0.05
0.89
Shared-cluster accuracy
temporal split (train ≤ 16 hpf)
random 80/20 split
majority (time split)